CS/SE 3RA3

Software Requirements and Security Considerations (Fall 2024)

McMaster University
ENGINEERING
Computing & Software

INSTRUCTORS

- **Sébastien Mosser**, PEng, PhD.
    - *Associate Professor, Dept. of Computing and Software* (mossers@mcmaster.ca)
    - Office hours: by appointment, via MS Teams

- **Teaching assistants**:
    - Cyruss Allen Amante *(program: B.Eng in Software Engineering)*
    - Stepan Bryantsev *(program: MASc in Software Engineering)*
    - Aidan Goodyer *(program: B.Eng in Software Engineering)*
    - Muhamad Jawad *(program: B.Eng in Software Engineering)*
    - Tiago de Moraes Machado *(program: PhD in Software Engineering)*
    - Carlos Zegarra *(program: PhD in Software Engineering)*

GENERAL INFORMATION

CALENDAR DESCRIPTION

Software requirements gathering. Critical systems requirements gathering. Security requirements. Traceability of requirements. Verification, validation, and documentation techniques. Software requirements quality attributes. Security policies. Measures for data confidentiality. Design principles that enhance security. Access control mechanisms.

LEARNING OBJECTIVES

- **Know and understand**:
    - The role of requirements engineering as part of software engineering;
    - The impact of requirements engineering on software developers& general public
    - How security impacts requirements
- **Be able to**:
    - Extract and model requirements out of imprecise customer descriptions
    - Organize requirements in a comprehensive and consistent document
    - Reflect on the role of security on requirement development.

COURSE PAGE

The course uses *Avenue To Learn* for pedagogical material and assignment deliveries.

- https://avenue.cllmcmaster.ca/d2l/home/630569

SCHEDULE

Slot	Monday	Tuesday	Wednesday	Thursday	Friday
08:30 09:20	Tutorial T01				
09:30 10:20		Tutorial T04			
15:30 16:20		Lecture C01	Lecture C01		Lecture C01
17:30 18:20			Tutorial T02	Tutorial T03	

- *For privacy reasons, check Mosaic for lecture and tutorial locations.*
- Attendance to lectures and tutorials is **mandatory**.
- The instructor reserves the right to cancel a scheduled lecture according to unexpected circumstances.
- Lectures and tutorials are designed as interactive sessions requiring the active participation of students and, as such, **will not be recorded**.

TEXTBOOK AND COURSE MATERIAL

No textbook is mandatory for this course. Slides from the lecture will be available on Avenue (but are not a textbook or lecture notes). The following books are "recommended" books that complement the course material.

The bibliography was curated to contain reference books from a state-of-practice point of view, freely available for students through McMaster Library:

- Handbook of Requirements and Business Analysis (Bertrand Meyer, 2022)
    - *https://mcmaster.primo.exlibrisgroup.com/permalink/01OCUL_MU/6thkva/cdi_askewsholts_vlebooks_9783031067396*
- Requirements Engineering, a Good Practice Guide (Sommerville & Sawyer, 1997)
    - *Available at Thodes Library*
- Software Requirements (Wiegers & Beatty, 2013)
    - *https://mcmaster.primo.exlibrisgroup.com/permalink/01OCUL_MU/deno1h/alma991029833629707371*
- UML@Classroom: An Introduction to Object-Oriented Modeling (Kappel et al, 2015)
    - *https://mcmaster.primo.exlibrisgroup.com/permalink/01OCUL_MU/deno1h/alma991029833629707371*

ONLINE COMPONENT & COMMUNICATION

- **The course uses Avenue as its main website** to make available lecture slides, tutorial material, and assignments.
- To store requirements, we will use the **GitHub classroom platform** so that students can start to build their portfolios.
- **We will use MS Teams as the one and only communication media** to support interaction between students and course staff (instructor and TAs).
    - The MS Teams channels and direct messages will be monitored during "regular" working hours (9 AM – 6 PM, weekdays only) to respect the staff's work-life balance.
    - We will try to answer questions promptly and within a maximum of 2 business days.
    - Messages that do not contain personal information **must** be posted publicly so that answers benefit all the students.
    - **Emails will not be answered.**

- **A communication ban is enforced at the end of each assignment.** All questions related to an assignment will **NOT** be answered within 72 hours before its deadline or during reading week.

COURSE CONTENTS

OUTLINE

The course will (tentatively) cover the following topics. Lectures can be “regular” lectures (L), “case studies” (CS), or “workshops” (W).

#Week	Lecture date	Topic	Type
#36	Tuesday	Opening Lecture: Welcome to 3RA3!	*L*
	Wednesday	The case for Good Requirements	*L*
	Friday	Artists & Specifiers	*W*
#37	Tuesday	Single Statement of Needs	*L*
	Wednesday	Identifying Stakeholders	*L*
	Friday	Persona Modelling	*W*
#38	Tuesday	Introduction to Goal Modelling	*L*
	Wednesday	Understanding Customers’ needs	*Guest*
	Friday	Decomposing Goals	*L*
#39	Tuesday	Overview of Risk Management	*L*
	Wednesday	Communicating & Working with Stakeholders	*L*
	Friday	Requirements for Oppression *(Ethical discussion)*	*L*
#40	Tuesday	Requirements Engineering Activities	*L*
	Wednesday	Refining/Organizing Requirements	*W*
	Friday	Why should we care? The case of Ageing Population	*CS*
#41	Tuesday	Role of Modelling in Requirements Engineering	*L*
	Wednesday	Use case models	*L*
	Friday	Scenario Modelling	*L*
#42	*READING WEEK – No Lectures*		
#43	**Tuesday**	**Midterm exam**	
	Wednesday	Domain modelling: Classes & Components	*L*
	Friday	Behavioural modelling: States & Activities	*L*
#44	Tuesday Wednesday Friday	The CONGO Case Study *(Modelling requirements at scale for an e-commerce platform)*	*CS*
#45	Tuesday	Principles of Agile Requirements	*L*
	Wednesday	User Stories in Action	*W*
	Friday	Acceptance Criteria & Definition of Done	*W*
#46	Tuesday	Evaluating the Quality of Requirements	*L*
	Wednesday	Digging deeper on non-functional requirements	*L*
	Friday	AI & Requirements Engineering: Extracting concepts from Text	*CS*
#47	Tuesday	The need for Security Policies	*L*
	Wednesday Friday	Security Policies in Practice	*CS*
#48	Tuesday	Overview of Cryptography and Steganography	*L*
	Wednesday	Privacy for mental-health related applications	*CS*
	Friday	Closing Lecture: Exam Walkthrough	
#49	Tuesday	*Buffer slot, if needed*	
	Wednesday	*Buffer slot, if needed*	

- **The instructor reserves the right to modify this tentative planning during the term.**
    - *If any modification becomes necessary, reasonable notice and communication with the students will be given with an explanation.*

GRADING & EVALUATION

The course contains three (3) graded elements: a term-long project, a midterm exam, and a final exam. The final (literal) grade is computed as a weighted average of the different assignments, as described in the table below.

ID	Assignment	Kind	Weight	Start date	Delivery date
P	ACME Project	*Team*	40%	03/09/24	05/12/24
M	Midterm (MCQ)	*Individual*	15%	22/10/2024, 15:30	
F	Final exam	*Individual*	45%	*To be announced by registrar*	

- The project is delivered thanks to Avenue (deadline is 23:59 CEST).
    - *Activity on the project is assessed with the GitHub repository.*
    - *The project is conducted by teams of three students*
    - *The project is graded on a [0,100] scale. See the assignment for detailed rubrics.*
    - ***There is no credit for deliveries handed in after the deadline: the train has already left the station.***
- The instructor reserves the right to modify this planning during the term.
    - *If any modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.*

OPPORTUNITY FOR EXTRA CREDIT

3RA3 offers two possibilities for extra credit this term:

1. **Individual**: Students can submit a "Meaningful & Memorable" self-reflection each week through a survey on Avenue.
    a. Survey answers will be assessed individually at the end of the course
    b. Participation **can add up to +1 to the 3RA3 final grade**.
2. **Cohort**: The participation rate in the *Student Course Experience Survey* **influences the rounding method** to compute the final grade.
    a. *> 60%: Midterm is rounded up.*
    b. *> 75%: Project assignment is rounded up.*
    c. *> 85%: Final exam is rounded up.*
    d. *> 95%: 3RA3 (weighted) grade is rounded up.*

POLICY ON GENERATIVE AI

Students are not permitted to use generative AI in this course. In alignment with McMaster's academic integrity policy, it "*shall be an offence knowingly to […] submit academic work for assessment that was purchased or acquired from another source*".

This includes work created by generative AI tools.

The policy also states, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." **Using Generative AI tools is a form of contract cheating**. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

MSAF POLICY

- **Self-reported MSAFs** are automatically accepted (no need to email) when submitted:
  - For the **midterm**, the exam is cancelled, and **the weight carries on to the final** (now weighted as 60% of the final grade).
  - *Windows of opportunities are optional deliveries and, as such, can't be MSAF-ed.*
  - *The term-long project is weighing more than 35% and can't be MSAF-ed.*
- **Administrative MSAFs** impact on project work will be discussed with the student after being accepted by the Faculty.

ACADEMIC INTEGRITY & PLAGIARISM[1]

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. The academic credentials you earn are rooted in honesty and academic integrity principles.

Individual assignments must be solved by one person only, **any** outside source (this includes asking other people or using any books or information found on the web) **has** to be documented. In compliance with the senate regulations on academic integrity, I remind you that: **People who let other people copy are as guilty as the ones who copy.** You can consult outside sources, meaning textbooks or the web, but **any** use of an outside source **must** be documented. **Code plagiarism detection tools will be used systematically**.

Similarly, group assignments must be solved by members of that group only, and the University policies apply as well (see "*Advisory Statements*" below).

MAPPING TO GRADUATE ATTRIBUTES

Attribute	Indicator		Evaluation		
			P	M	F
Knowledge Base	1.4	*Competence in specialized engineering knowledge*	✓	✓	✓
Problem analysis	2.1	*Identifies and states reasonable assumptions and suitable engineering fundamentals, before proposing a solution path to a problem*	✓		✓
	2.2	*Proposes problem solutions supported by substantiated reasoning, recognizing the limitations of the solutions*	✓		✓
Investigation	3.1	*Selects appropriately from relevant knowledge base to plan appropriate data collection methods and analysis strategies*	✓	✓	
	3.2	*Synthesizes the results of an investigation to reach valid conclusions*	✓		✓
Design	4.1	*Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions*	✓	✓	✓
	4.2	*Explores a breadth of potential solutions, considering their benefits and trade-offs as they relate to the project requirements*	✓		
Use of engineering tools	5.1	*Evaluates engineering tools, identifies their limitations, and selects, adapts, or extends them appropriately*	✓	✓	✓

[1] *Adapted from Dr. Jacques Carette outline for SFWRENG 3GB3 – Fall 2021*

Communication skills	7.1	*Demonstrates comprehension of technical and non-technical instructions and questions*	✓		
	7.2	*Composes an effective written document for the intended audience*	✓		
	~~7.3~~	~~*Composes and delivers an effective oral presentation for the intended audience*~~			
Economics and project management	11.1	*Applies economic principles in decision making*	✓		✓

ADVISORY STATEMENTS

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, *e.g.* the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "*Grade of F assigned for academic dishonesty*"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the **Academic Integrity Policy**, located at https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/

The following illustrates only three forms of academic dishonesty:

- plagiarism, *e.g.* the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE COMPONENT

Some courses may use on-line elements (*e.g.* e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the Policy on Requests for Relief for Missed Academic Term Work.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.